

CSE 422: ARTIFICIAL INTELLIGENCE

PROJECT TITLE: Tree Sterility Classification using ML

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1. Introduction

The aim of this project is to study and observe machine learning models that can classify whether a tree is sterile or non-sterile based on various types features. By analyzing this dataset, we can identify which factors or which features most influence sterility and provide best and reliable models.

The motivation behind this project is to apply supervised learning techniques to the real-world Tree Sterility Dataset, compare the results with 5 models that works best for classification problems— Neural Networks, Decision Tree, Logistic Regression, Random Forest, and KNN models and evaluate their effectiveness using multiple performance metrics.

2. Dataset Description

Dataset Overview

- **Dataset Name:** Tree Sterility Dataset
- **Number of Features:** 23, excluding the target feature- "Sterile"
- **Type of Problem:** Classification – since the target variable "Sterile" represents discrete category ("Non-Sterile" or "Sterile)
- **Number of Data Points:**
  Total datapoints (rows): 2783
  Total features (columns): 23
  Total data in the whole table: 64,009 entries

- **Feature Types:**
  1. **Quantitative:** "No", "Plot", "Light_ISF", "Core", "PlantDate", "AMF", "EMF", "Phenolics", "Lignin", "NSC", "Census", "Time", "Event"
  2. **Categorical:** "Subplot", "Species", "Light_Cat", "Soil", "Sterile", "Conspecific", "Myco", "SoilMyco", "Harvest", "Alive".

- **Need for Encoding:**
  Categorical features cannot be directly used in most ML models. For instance:
  Neural Networks take only numerical inputs so categorical variables must be enoded before feeding to the network.

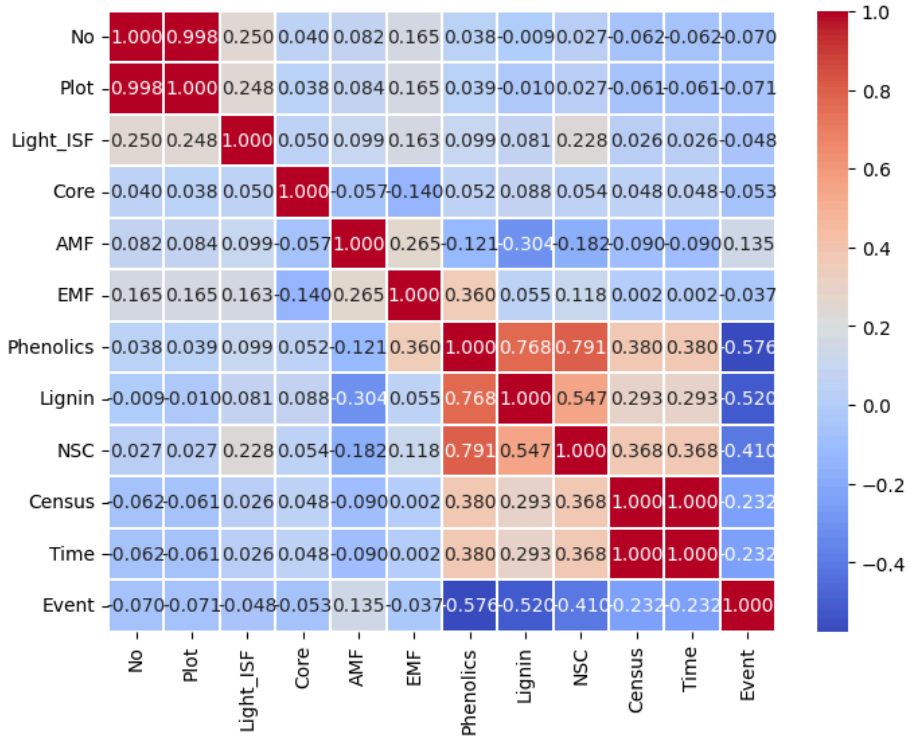
Logistic Regression is a mathematical model that computes weighted sum of numeric inputs which is why categorical values cannot be directly handled.

Therefore, encoding is required. There are two types of encoding and we will be using them:

1. Label Encoding
2. One-Hot Encoding

Correlation of Features (with Heatmap)

- A heatmap (Seaborn) is plotted to show correlation between features and target variable. It measure the linear relationship between two continuous (numeric) variables. Perfect Positive Linear Correlation are those close to +1.0, Perfect Negative Linear Correlation are those close to -1 while 0 indicates no linear correlation.



Observation:

Using Correlation Matrix, the correlation relationship between two features can be observed and studied.

For example:

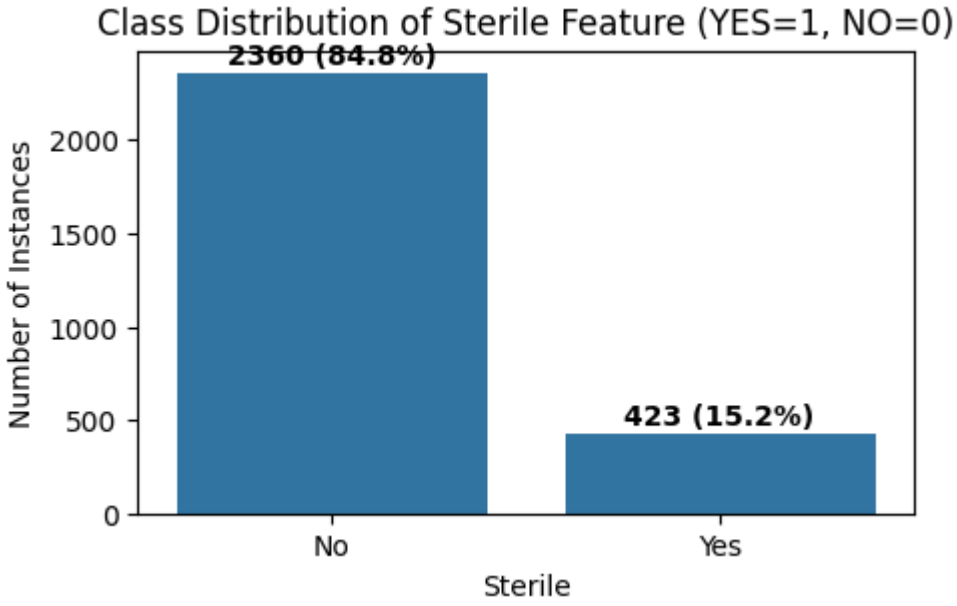
There is strong positive linear correlation between:
"Phenolics" and "NSC", "Phenolics" and "Lignin", and "Lignin" and "NSC"
There is strong negative linear correlation between:
"Phenolics" and "Event", "Lignin" and "Event", "NSC" and "Event"
While the others are in the Low / Weak Correlations Category.

Additionally, there is multicollinearity issues which means having too high correlations. It can be seen in the matrix, colored in red. "Census" and "Time" are identical and "No" and "Plot" are almost identical. Keeping both introduces redundancy, so should be dropped.

- **Imbalanced Dataset**

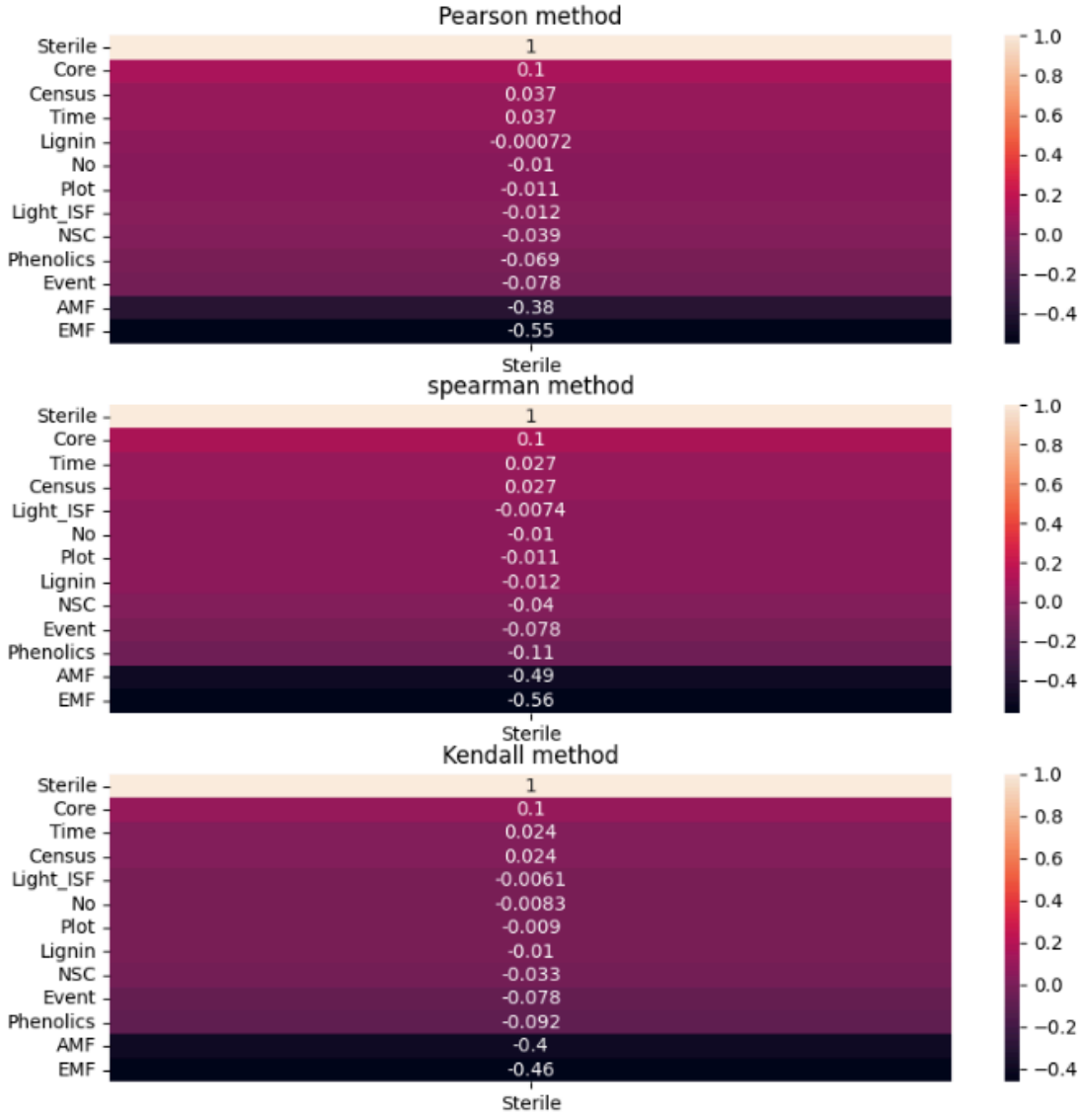
  For output feature "Sterile", all unique classes do not have an equal number of instances.
  Class 0 (Non-Sterile) – 2360 instances (85% approx)
  Class 1 (Sterile) – 423 instances (15% approx)

  This shows a clear class imbalance where the majority of trees are not sterile.

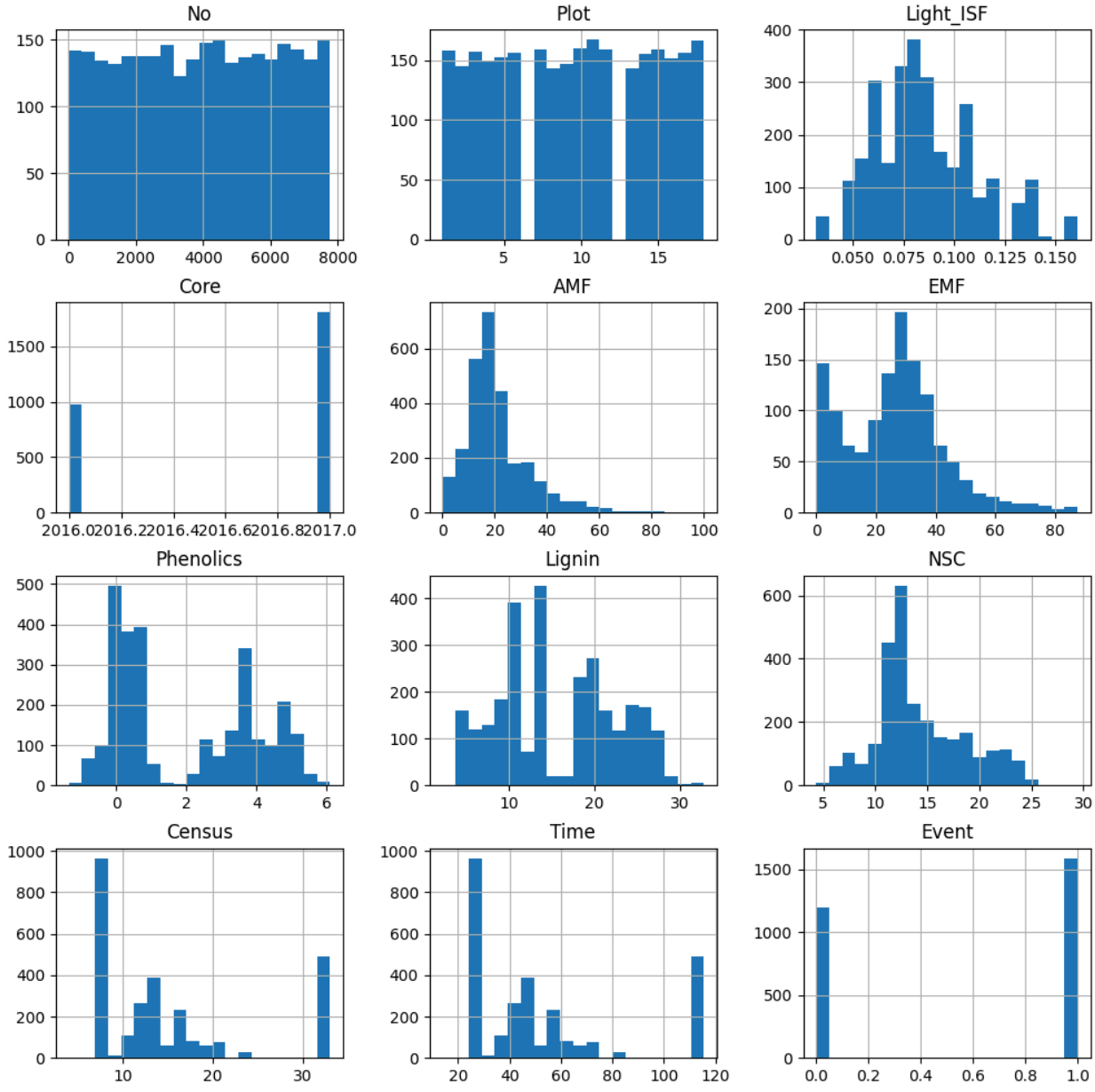


ADDITIONAL DATA ANALYSIS: EDA

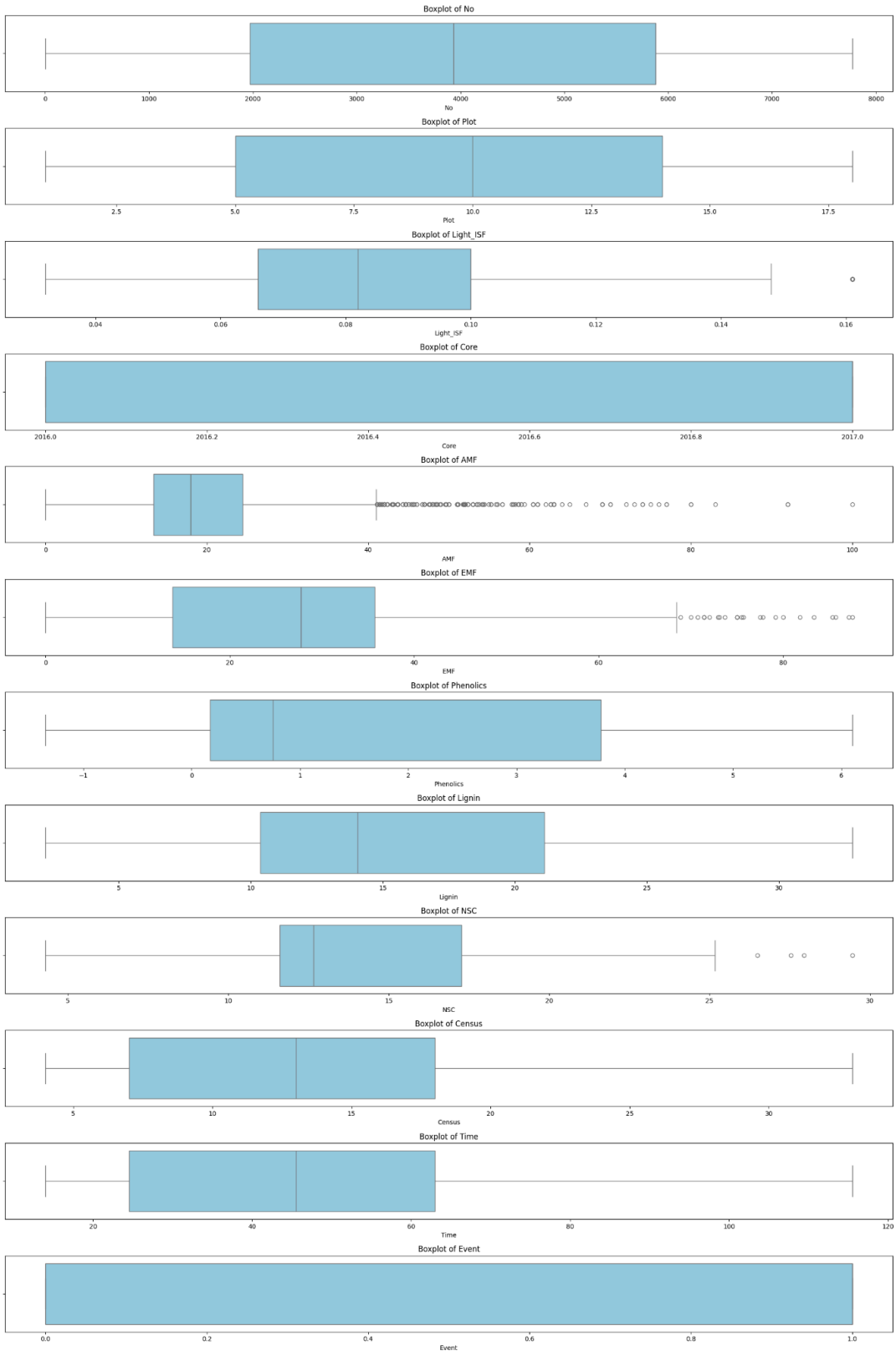
- Correlation Plot Between Numerical Features and Target



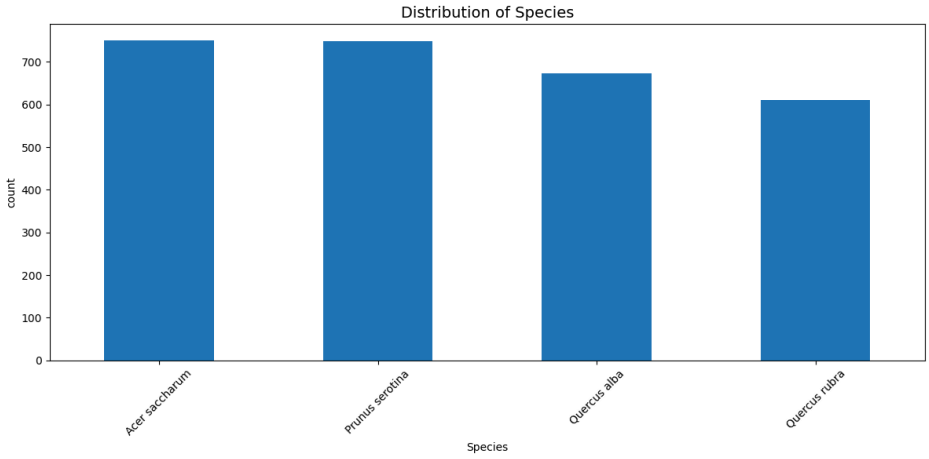
- **HISTOGRAM AND BOX PLOTS**

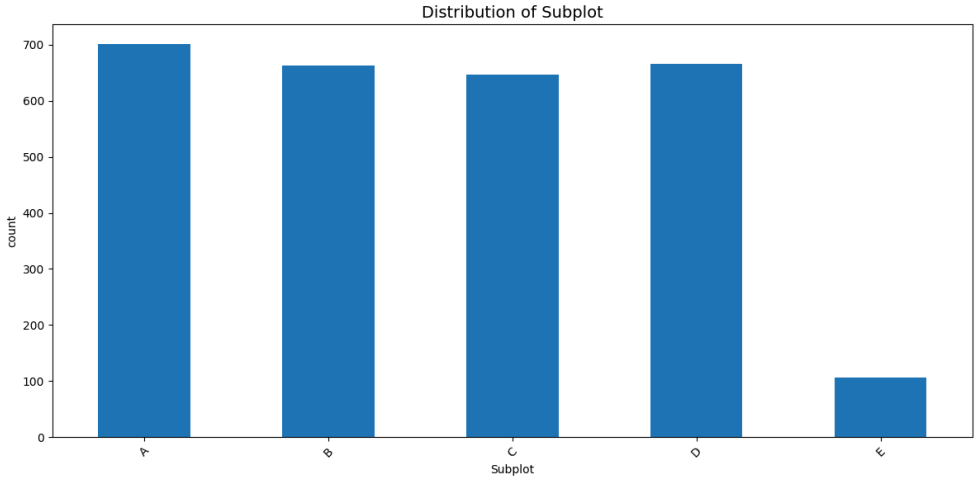


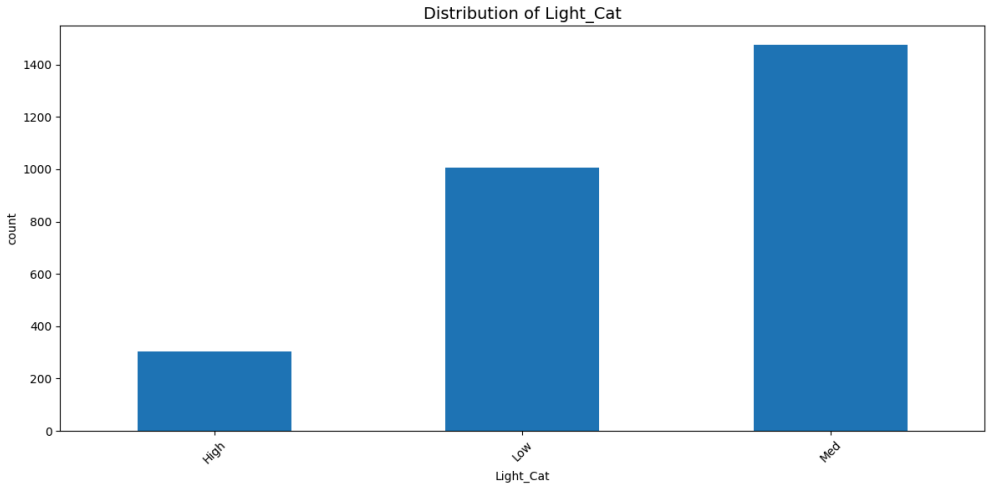
BOXPLOT ANALYSIS

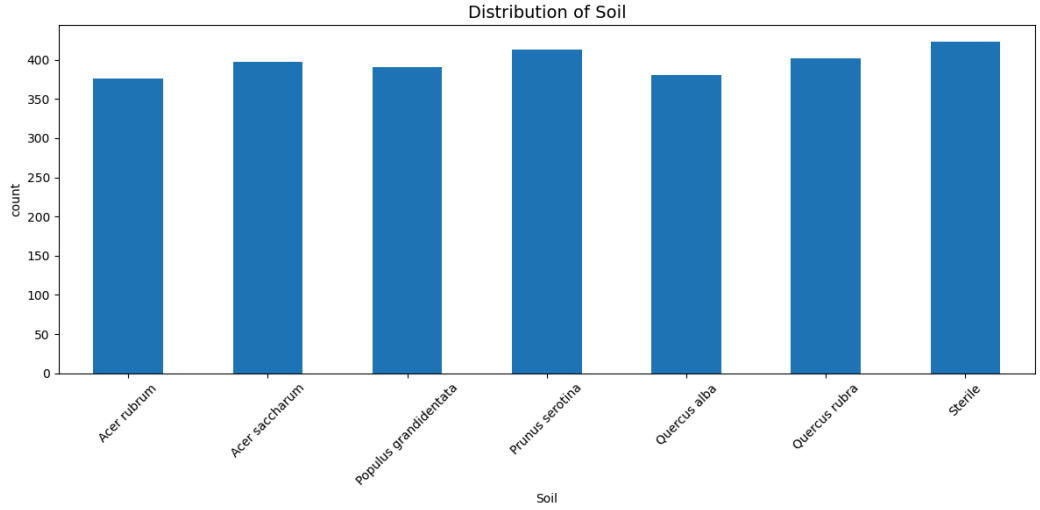


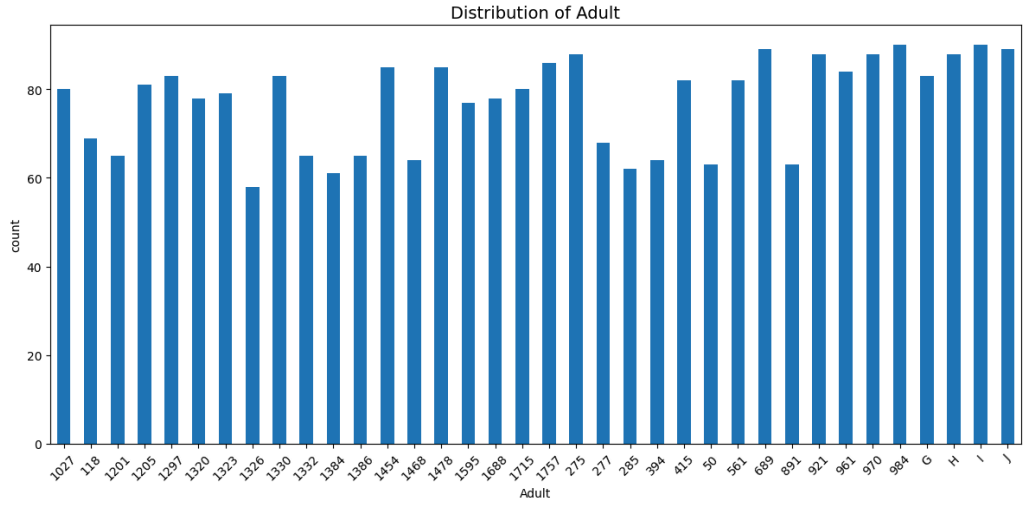
BARPLOT OF UNIQUE VALUE COUNTS IN EVERY CATEGORICAL FEATURES

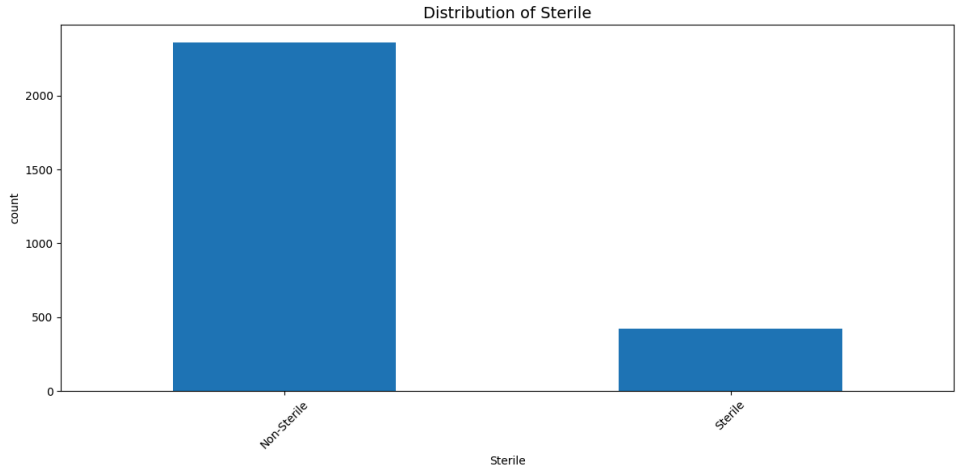


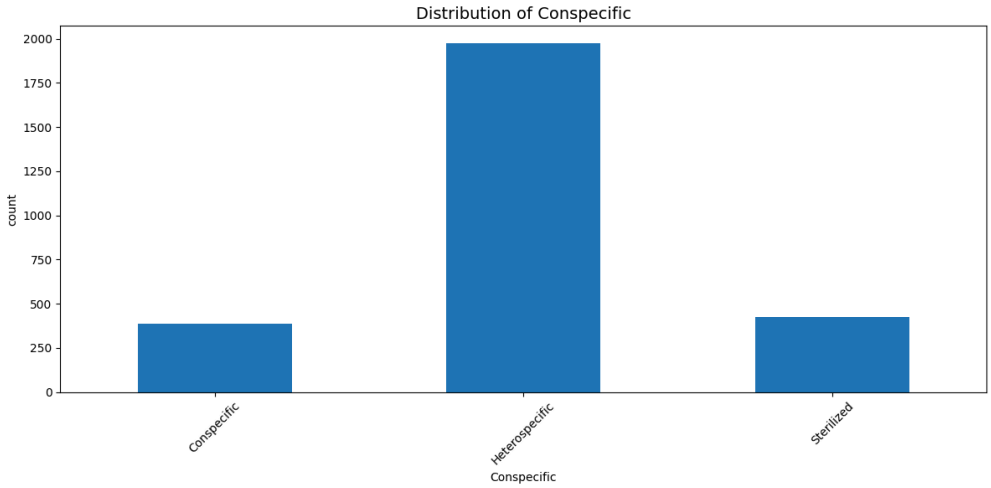


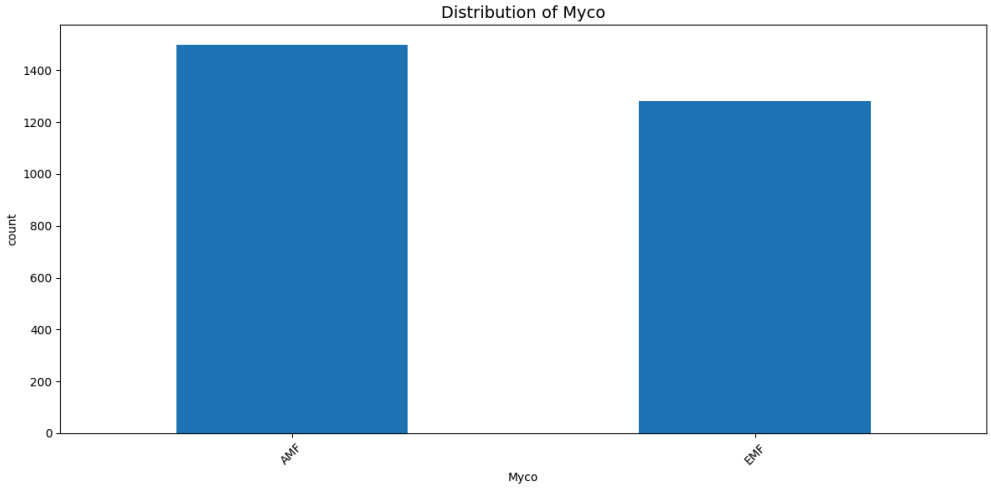


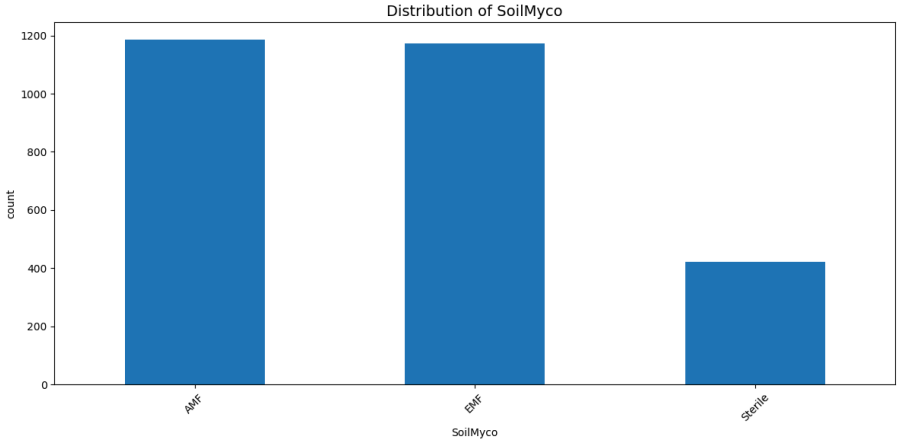

Distribution of Soil
count
0
50
100
150
200
250
300
350
400
Acer rubrum
Acer saccharum
Populus grandidentata
Prunus serotina
Quercus alba
Quercus rubra
Sterile
Soil

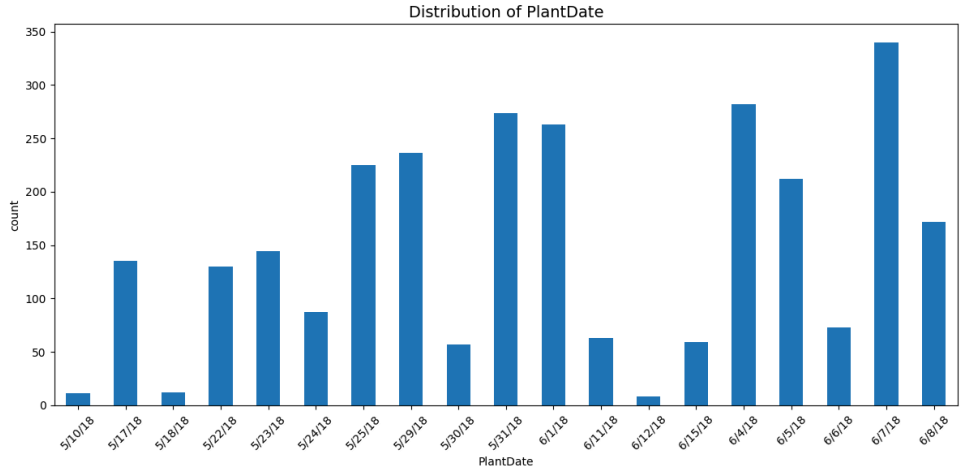

Distribution of Adult
count
0
20
40
60
80
1027
118
1201
1205
1297
1320
1323
1326
1330
1332
1384
1386
1454
1468
1478
1595
1688
1715
1757
275
277
285
394
415
50
561
689
891
921
961
970
984
G
H
I
J
Adult

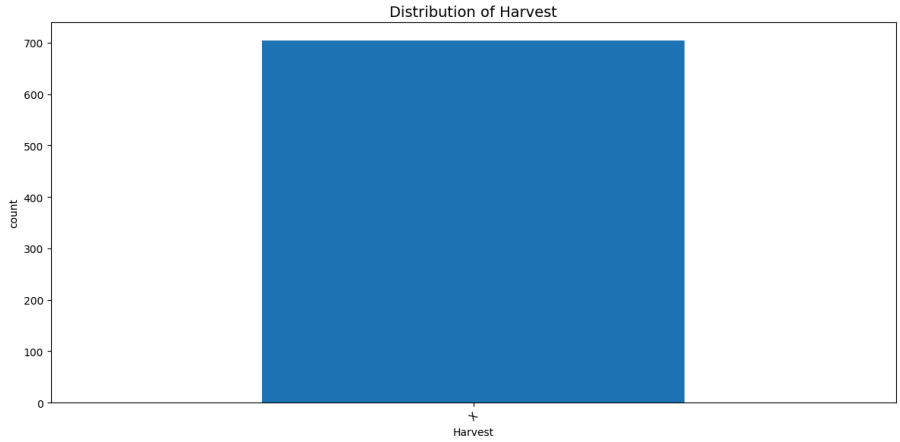

Distribution of Sterile
count
0
500
1000
1500
2000
Non-Sterile
Sterile
Sterile

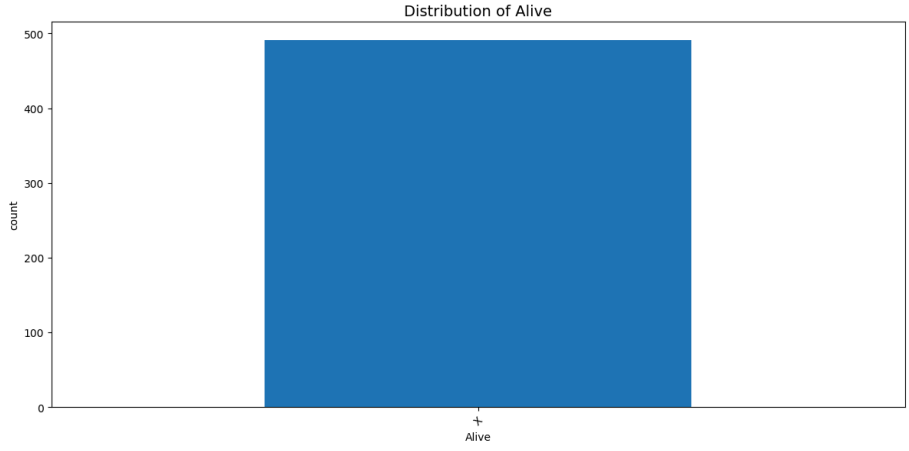

Distribution of Conspecific
count
2000
1750
1500
1250
1000
750
500
250
0
Conspecific
Heterospecific
Sterilized
Conspecific


Distribution of Myco
count
1400
1200
1000
800
600
400
200
0
AMF
EMF
Myco

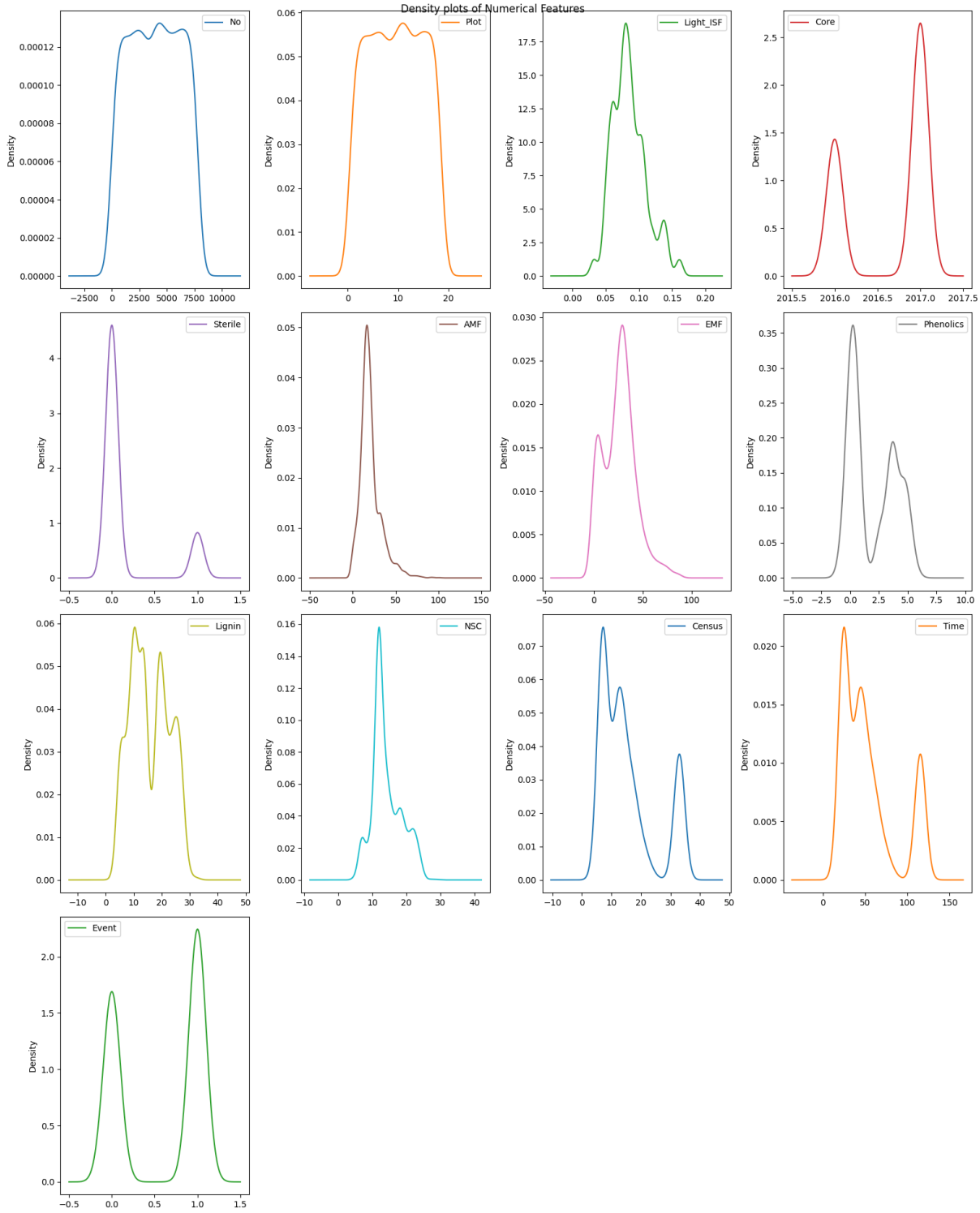

Distribution of SoilMyco
count
1200
1000
800
600
400
200
0
AMF
EMF
Sterile
SoilMyco


Distribution of PlantDate
count
PlantDate
5/10/18
5/17/18
5/18/18
5/22/18
5/23/18
5/24/18
5/25/18
5/29/18
5/30/18
5/31/18
6/1/18
6/11/18
6/12/18
6/15/18
6/4/18
6/5/18
6/6/18
6/7/18
6/8/18


Distribution of Harvest
count
Harvest


Distribution of Alive
count
Alive

DENSITY PLOTS OF NUMERICAL FEATURES:



3. DATA PRE-PROCESSING

Faults:

1. Null/Missing Values:

The total number of NAN values for “Harvest” is 2079 and for “Alive” is 2292.
Too many NAN values in the dataset won’t produce the accurate result.
Solution: dropped these 2 columns.

2. Redundant Feature:

“SoilMyco” feature is the same as “Myco” with additional wrong “Sterile” inputs.
Solution: dropped this column

3. Redundancy Observed From Correlation Matrix:

“Plot” and “Census” becomes redundant and unnecessary due to very high correlation rate.
Solution: dropped these columns

4. Wrong imputes for numerical features:

“Adult” feature contains mostly numeric values. However, there are some wrong letters inputted.
Solution: imputed mean of all the numerical number using SimpleImputer to replace the wrongly inputted letters.

5. NULL VALUES OF “EMF”

EMF feature could be important for our dataset analysis, however, there are many null values.
Solution: Imputed values of mean of all the numerical values using SimpleImputer

6. Wrong categorical values in “Conspecific” features

Unique values should have been Heterospecific and Conspecific but there were “Sterilized” imputed which is wrong.
Solution: replaced all the “Sterilized” with mode of Conspecific feature

FEATURE SCALING:

Numerical values were scaled using Robust Scalar since it works best for handling outliers which is present in this dataset. It works well for models sensitive to feature scale like Neural Networks and Logistic Regression.

4. DATA SPLITTING:

Train-Test Split: 70% training, 30% testing.

Stratified Split: Used since target (Sterile) has **85% "No"** and **15% "Yes"/**
Stratified split guarantees both sets stay close to 85%/15%.

5. Model Training & Testing

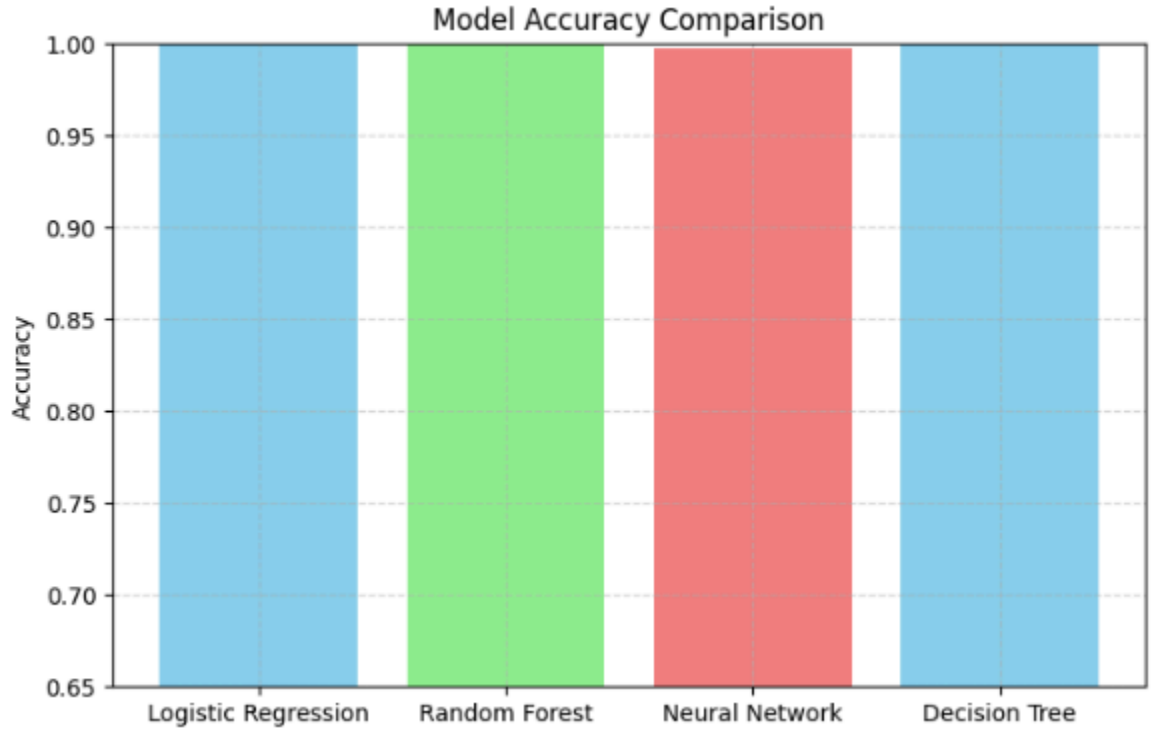
Models Applied

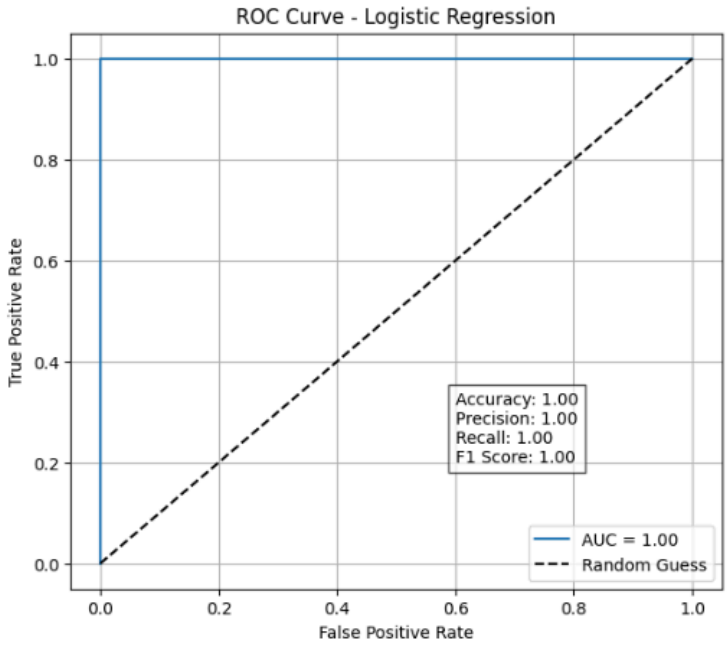
1. **Decision Tree Classifier**
    - Works well with categorical data.
    - Easy to interpret feature importance.
2. **Logistic Regression**
    - Suitable for binary classification.
    - Requires scaling and encoding.
3. **Neural Network (MLP Classifier)**
    - Implemented with sklearn.neural_network.MLPClassifier.

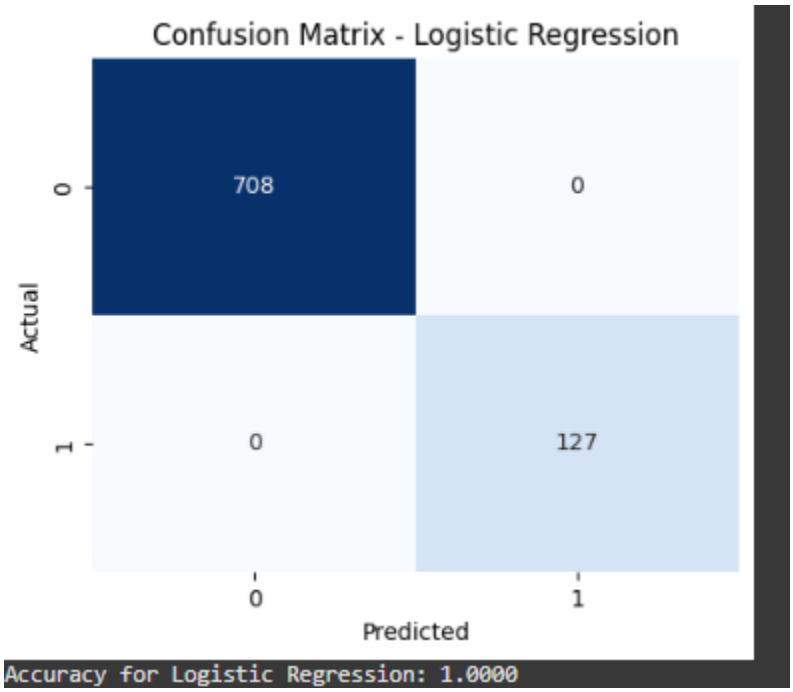
4. Random Forest

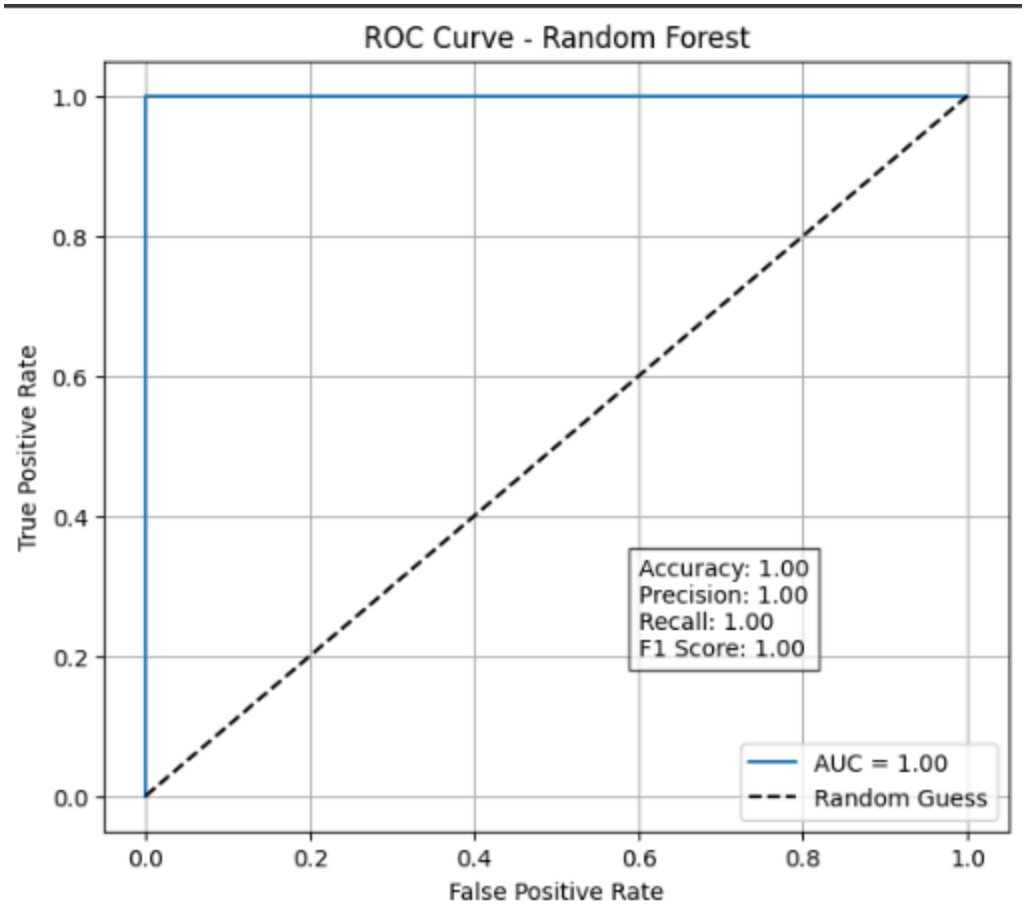
- Works with both categorical and numerical features
- Robust to outliers
- Good for Imbalanced Datasets

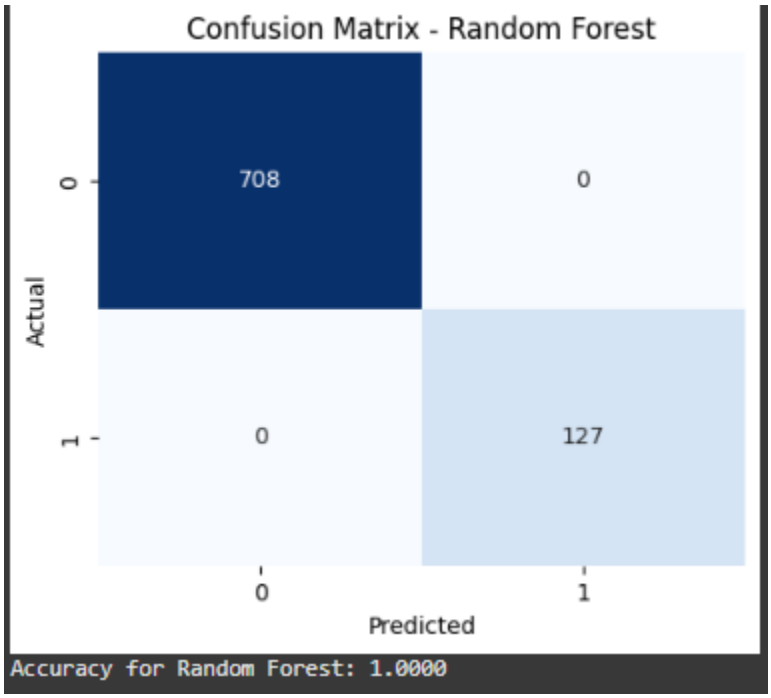
6. Model Selection / Comparison Analysis:

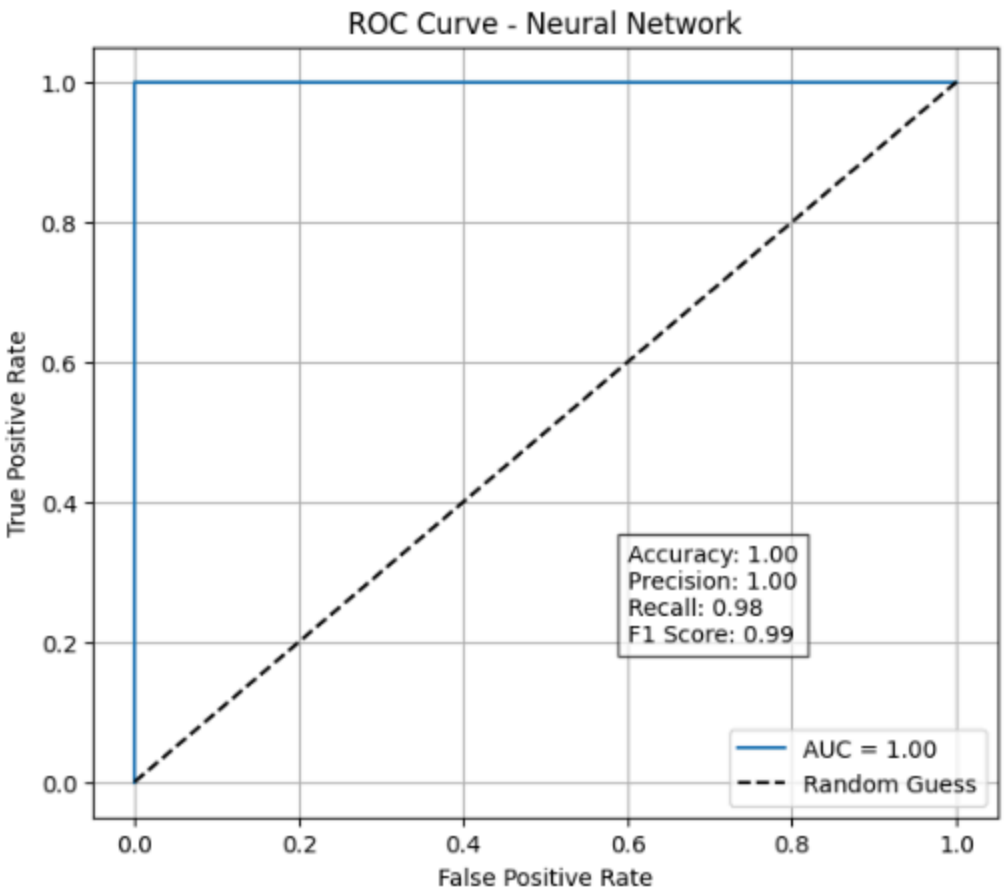


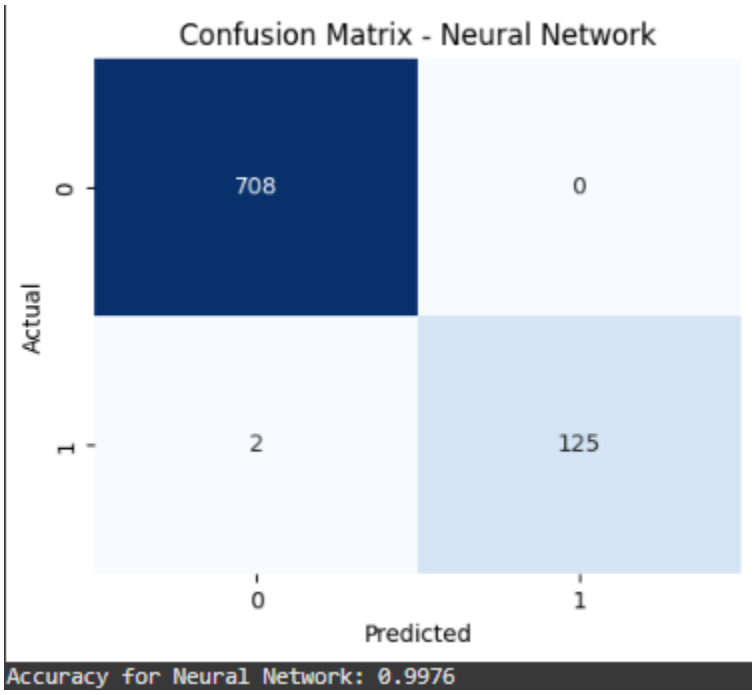

ROC Curve - Logistic Regression
True Positive Rate
False Positive Rate
Accuracy: 1.00
Precision: 1.00
Recall: 1.00
F1 Score: 1.00
AUC = 1.00
Random Guess


Confusion Matrix - Logistic Regression
708
0
0
127
Actual
Predicted
Accuracy for Logistic Regression: 1.0000

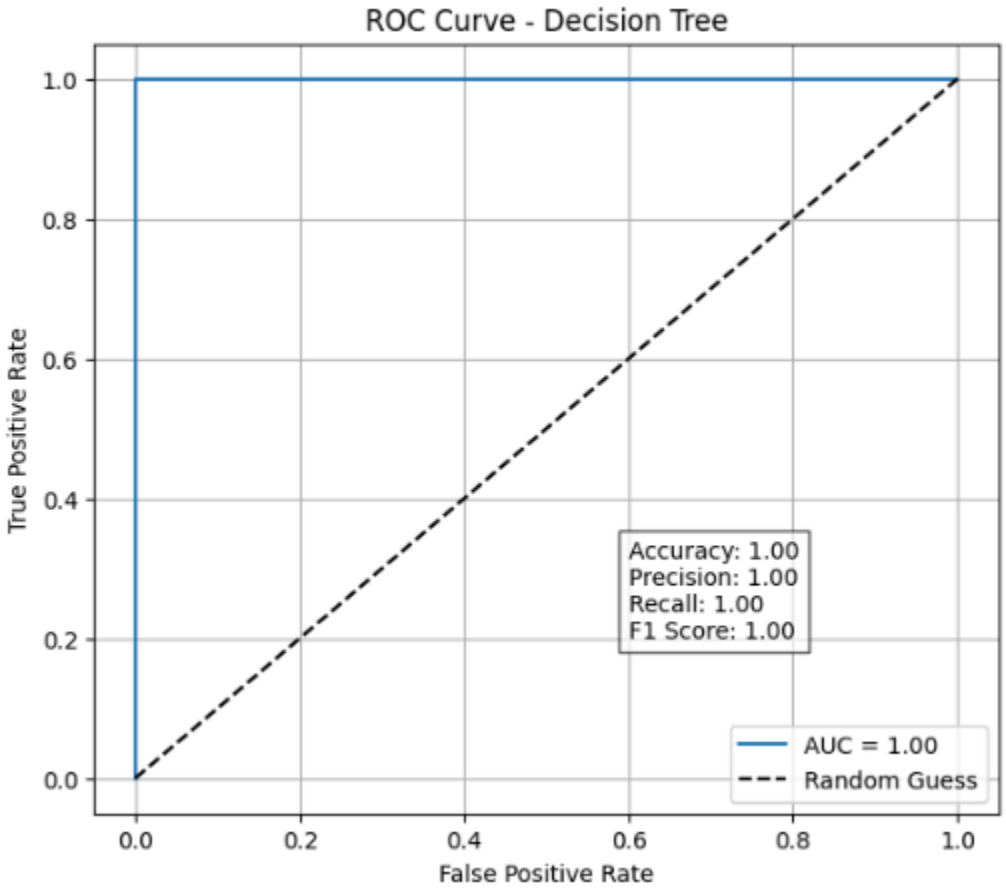

ROC Curve - Random Forest
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Accuracy: 1.00
Precision: 1.00
Recall: 1.00
F1 Score: 1.00
AUC = 1.00
Random Guess

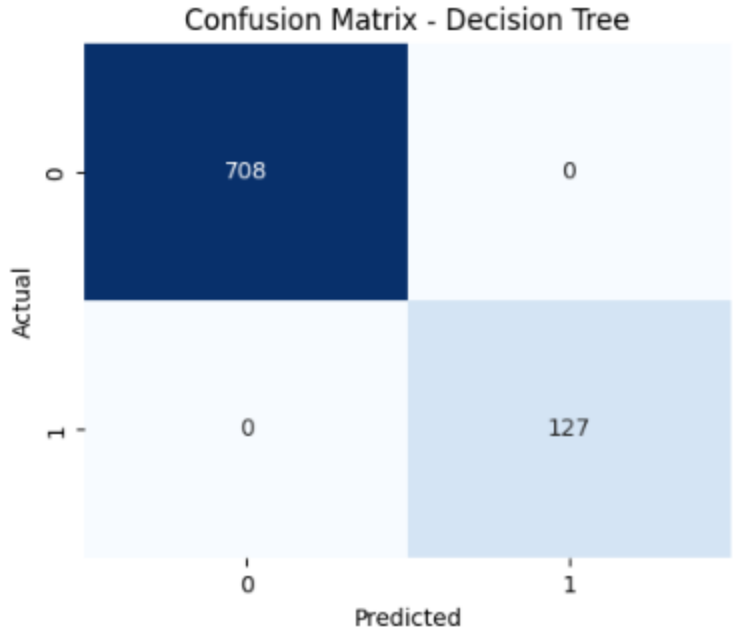

Confusion Matrix - Random Forest
708
0
0
127
Actual
Predicted
0
1
Accuracy for Random Forest: 1.0000






```
Accuracy for Neural Network: 0.9976
```

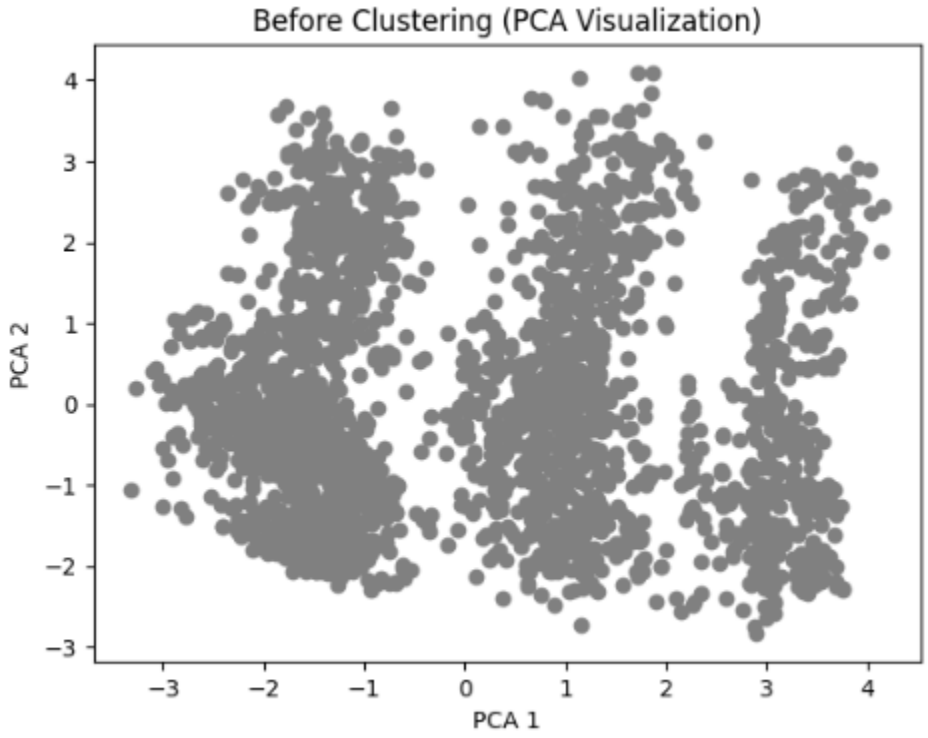

ROC Curve - Decision Tree
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Accuracy: 1.00
Precision: 1.00
Recall: 1.00
F1 Score: 1.00
AUC = 1.00
Random Guess

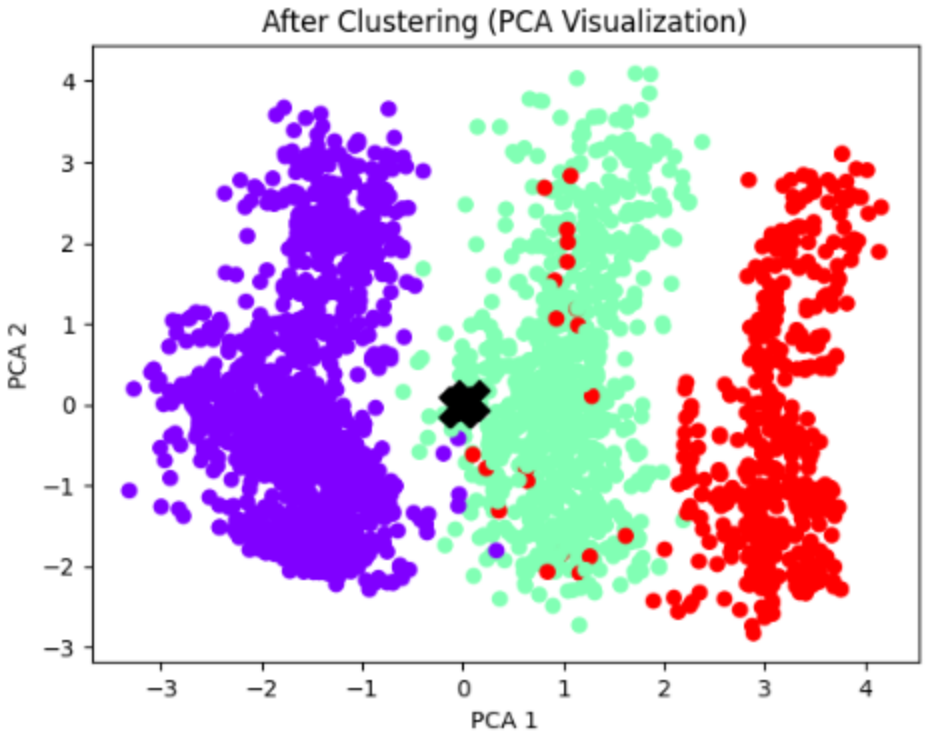


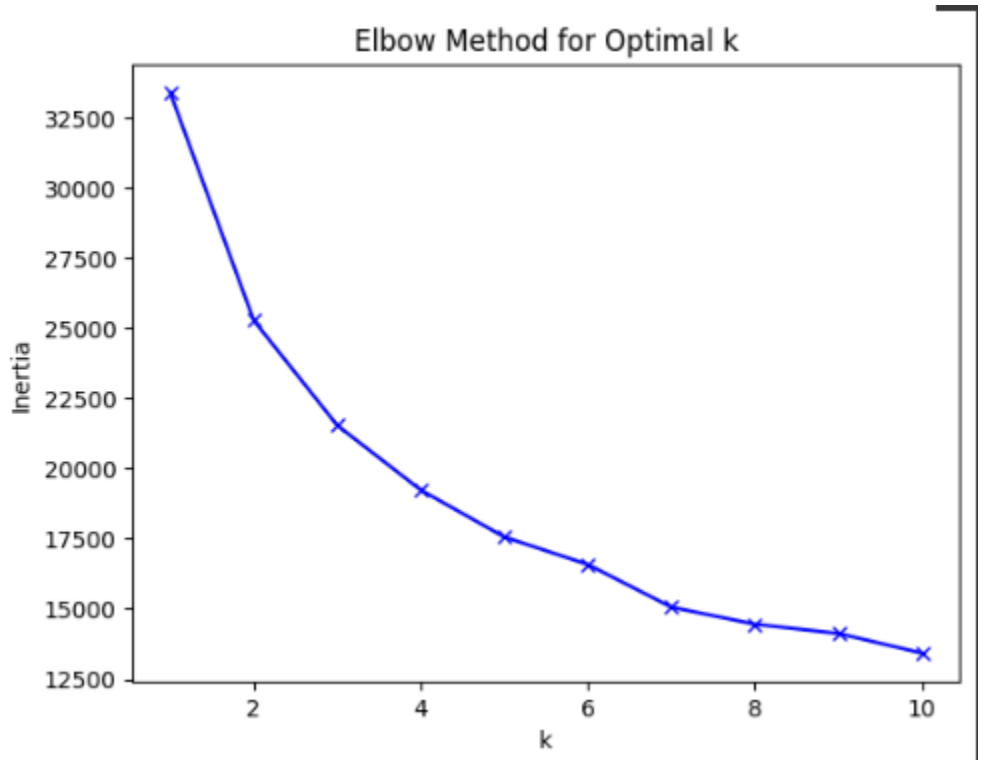

```
Accuracy for Decision Tree: 1.0000
```

8. TREATING THE DATASET AS UNSUPERVISED:

Applying Kmeans Clustering:




After Clustering (PCA Visualization)
PCA 2
PCA 1
4
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3
4


Elbow Method for Optimal k
Inertia
k
32500
30000
27500
25000
22500
20000
17500
15000
12500
2
4
6
8
10

9. Conclusion:

In this project, several machine leaving models were applied to predict tree sterility including Logistic Regression, Decision Tree, Random Forest, and Neural Network:

- All models achieved high accuracy, which indicates that the dataset is quite predictive for the target variable.
- Logistic Regression, Random Forest, and Decision Tree performed slightly better than the Neural Networks models
- The fact that the models reached similar results indicates that the problem is inherently easy to classify with this dataset
- Features such as Phenolics, Ligin, and NSC appear to have strong correlation with sterility making the classification task straightforward.
- Good Preprocessing: Encoding, scaling, stratified splitting, and handling imbalance ensured clean data and fair evaluation, boosting performance.

- **CHALLENGES FACED:**

Main challenged faced was during data pre-processing as there were many null and missing values to handle. Moreover, there were wrong variables inputting and there were redundancy features to handle as well.